

Assignment 7

Chris Conlon

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Econometrics I
NYU Stern

Professor Chris Conlon
Email: cconlon@stern.nyu.edu

Instructions. Two problems on binary choice and selection (Session 12). R with `fixest` (`feglm` for logit and probit) or Python with `pyfixest`. Turn in a PDF with your results and reasoning, and your code.

1 Mortgage denials in Boston

The `Hmda` data (`Ecdat` package in R; also in the course repository) contain 2,380 mortgage applications from the 1990 Boston HMDA study (Munnell et al. 1996). The outcome is `deny`. Use the specification from the Session 12 notebook: `black`, debt-to-income (`dir`), housing-expense-to-income (`hir`), loan-to-value indicators for medium ($0.8 \leq \text{lvr} \leq 0.95$) and high (> 0.95), consumer and mortgage credit scores (`ccs`, `mcs`), public bad credit record (`pbcr`), and denied mortgage insurance (`dmi`).

- (a) Report the denial rate for Black and for white applicants. Then estimate the linear probability model, the probit, and the logit with the covariates above. Present the three in one table with robust standard errors for the LPM.
- (b) The logit coefficient on `black` is roughly 1.8 times the probit coefficient. Why is that the expected ratio, and why does it not mean the models disagree?
- (c) Compute the effect of `black` on the denial probability three ways: the LPM coefficient; the logit *average* marginal effect (average over applicants of the predicted probability with `black` set to 1 minus set to 0); and the logit effect *at the means* of the covariates. Using the shape of the logistic CDF, explain when the last two differ (where would the “average applicant” have to sit for the effect at the means to be much smaller than the average effect?) and why they are close in these data.

- (d) Compare your estimates in (c) to the raw gap in (a). Two sentences: what accounts for the difference, and what would the coefficient on `black` have to be interpreted as for this regression to measure discrimination? (Munnell et al. argue one side; think about what an omitted variable would need to look like.)
- (e) What share of the LPM's fitted probabilities lie outside $[0, 1]$? Which applicants are they?
- (f) Classification. Using the logit's predicted probabilities, build the confusion matrix at a cutoff of 0.5 and at 0.2. Report accuracy, sensitivity (share of denials caught), and specificity. Note that always predicting "approve" has accuracy 0.88. One paragraph: which cutoff would a lender's compliance officer prefer, which would a loan officer prefer, and why is "accuracy" the wrong summary for a rare outcome?

2 Interest rates you only see for approved loans

Return to the loan officers of Assignment 4, but now the outcome is the *interest rate* on the loan, which exists only if the loan is approved. 4,000 applications, each randomly assigned to one of 40 officers with leniency $\ell_j \sim \mathcal{N}(0, 0.6^2)$:

$$\begin{aligned} \text{approve}_i &= \mathbf{1}\{0.5 \text{score}_i + \ell_{j(i)} + u_i > 0\} \\ \text{rate}_i &= 6 - 0.8 \text{score}_i + 0.5 e_i, \quad \text{observed only if approved,} \end{aligned}$$

with $\text{score}_i \sim \mathcal{N}(0, 1)$ and (u_i, e_i) standard normal with correlation $\rho = 0.6$: applicants the officer is inclined to approve for reasons you do not see are also charged more. The question is the effect of the credit score on the rate offered, -0.8 .

- (a) Simulate. Regress rate on score using approved loans only. Sign and explain the bias before you look at the number (Session 12: what is $\mathbb{E}[e \mid \text{approved}, \text{score}]$, and how does it move with score?).
- (b) Heckman two-step: probit of approval on score and officer leniency; compute the inverse Mills ratio $\lambda(\hat{z}) = \phi(\hat{z})/\Phi(\hat{z})$ from the probit index; regress rate on score and $\hat{\lambda}$ in the approved sample. Compare the score coefficient to -0.8 and the coefficient on $\hat{\lambda}$ to $\rho\sigma_e = 0.3$.
- (c) Redo (b) with a probit on score *alone*, no leniency. Report the coefficient on score and its standard error, and the correlation between $\hat{\lambda}$ and score in the approved sample. One paragraph: what is the *exclusion restriction* in this problem, why is officer leniency a good one, and what happens to the two-step estimator without it?

- (d) Write the full-information likelihood for one observation (a probability for rejected applications; a density times a conditional probability for approved ones) and maximize it with `optim` (or `sampleSelection::selection` in R). Compare to (b).
- (e) Managerial summary, two sentences: a fintech competitor proposes to price loans using the OLS from (a), estimated on the incumbent's approved book. What will go wrong when they lend to applicants the incumbent would have rejected?